GG-N-211b
August 29, 1961

SUPERSEDING
Fed. Spec. GG-N-211a
August 26, 1958

FEDERAL SPECIFICATION

NEEDLE, SUTURE, (EYED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers various types, classes, points, and sizes of surgeons' suture needles.

1.2 Classification.

1.2.1 *Types.* Needles shall be of the following types, as specified (see 6.1):

Type I.—Surgeons' regular.
Type II.—Fine intestinal.
Type III.—Murphy intestinal.
Type IV.—Mayo intestinal.
Type V.—Mayo catgut.
Type VI.—Eye.
Type VII.—Kidney.
Type VIII.—Cervix.
Type IX.—Keith abdominal.
Type X.—King (skin).
Type XI.—Post mortem.
Type XII.—Martin's uterine.
Type XIII.—Ferguson abdominal.
Type XIV.—Gallie fascia.
Type XV.—French spring eye.
Type XVI.—Kalt corneal eye.
Type XVII.—Hagedorn.
Type XVIII.—Lane cleft palate.

1.2.2. *Classes.* Needles shall be of the following classes, as specified (see 6.1):

Class A.—⅜ Circle.
Class B.—½ Circle.
Class C.—Half curved.
Class D.—Straight.

1.2.3 *Points.* Needles shall be of the following points, as specified (see 6.1):

Point 1.—Cutting edge.
Point 2.—Trocar.
Point 3.—Taper.
Point 4.—Blunt.
Point 5.—Triangular.

1.2.4 *Sizes.* Sizes shall be as specified (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on data of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.

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Federal Standards:

Fed. Std. No.102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151.—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–116—Preservation, Methods of.

MIL–L–10547—Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Steel Products Manual, Stainless and Heat Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material. The needle shall be fabricated from American Iron and Steel Institute (AISI) 420 or 440A stainless steel. The composition of the needles shall be determined as specified in 4.4.4.

3.2 Style, design, and dimensions. Needles shall be of the type, class, point, and size, as specified (see 6.1). Style, design, and dimensions shall be in accordance with the applicable figures listed in table I. The point of the needle shall be surgically sharp, tapered, or otherwise formed as specified. The eye of the needle shall have smooth walls and shall be properly constructed for maximum efficiency. The "regular" eye only shall be furnished under this specification, except for type XV which shall be furnished with a "French spring eye". The suture grooves, where required, shall be formed on both sides of the needle.

TABLE I.—*Figures*

Figure	Type	Class	Point
1	I	A	1
2	I	B	1
3	I	C	1
4	II	B	3
5	III	B	3
6	IV	D	3
7	V	A	2
8	V	B	3
9	VI	A	1
10	VI	B	1
11	VI	B	3
12	VII	B	4
13	VII	C	4
14	VII	D	4
15	VIII	A	3
16	IX	D	5
17	X	A	1
18	XI	C	1
19	XII	B	1
20	XIII	B	3
21	XIV	B	3
22	XV	B	3
23	XVI	A	3
24	XVII	D	1
25	XVIII	B	1

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TABLE II.—*Tolerances in dimensions*

Material diameter (wire size)	Permissible variations (plus or minus)												
	A	B	C	D	E	F	G	H	K	L	T	R	U
Inch	*Inch*	*Inch*	*Inch*		*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
0.016 to 0.025	0.001	0.002	0.002	*	0.002	0.006	0.010	1/16	....	0.002	0.002	1/32	1/32
.028 to .038	.0015	.002	.002	*	.005	.010	....	....	....	.002	.002	1/16	1/32
.040 to .048	.0015	.005	.005	*	.010	.010	....	....	....	.005	.005	1/16	1/16
.050 to .076	.002	.005	.010	*	.010	1/64	1/64	1/64	1/64	.005	.005	1/16	1/16
.093 and over	.002	.005	.010	*	.010	1/64	....	....	....	.005	.005	1/16	1/16

* This measurement varies with the members of the trade; therefore, will not be held to a definite tolerance.

3.2.1 *Tolerances.* Tolerances for the various dimensions shall be as listed in table II according to the material diameter (wire size) used in the fabrication of the needle.

3.3 Passivation. As a final finishing operation, needles shall be properly passivated.

3.4 Corrosion-resistance.

3.4.1 *Resistance to boil test.* When tested as specified in 4.4.1.1, the needle, upon examination, shall not show corrosion on more than 2 percent of the needle surface.

3.4.2 *Resistance to copper sulfate test.* When tested as specified in 4.4.1.2, the needle shall show no plating of copper as a result of the test. Dulling of the polished surface, which may be caused by the copper sulfate solution, shall not be considered as noncompliance with this test.

3.5 Hardness. After proper stress relieving, the finished needles shall have a Rockwell hardness of C50 to C55 or its equivalent, as shown in the hardness conversion table in Federal Test Method Standard No. 151, when tested as specified in 4.4.2.

3.6 Bending. No breakage shall occur when tested in accordance with the bend test as specified in 4.4.3.

3.7 Finish. Needles shall have a smooth and bright finish. This finish may be accomplished by electropolishing.

3.8 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability. Needles shall be smoothly and symmetrically finished. Needles shall be free from filings, polisher's dirt, or other material which would necessitate cleaning prior to sterilization and use. Needles shall be sharp, not blunt (except where specified) or bent, and shall be free of feathered edges, burrs, or other defects, when examined under 4 to 7 power magnification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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TABLE III. *Sampling criteria*

Component	Characteristic	Req.	Test procedure	Sample unit	Insp. level	AQL (percent defective)
Needle	Boil test	3.4.1	4.4.1.1	1-Needle	L3*	**
	Copper sulfate test	3.4.2	4.4.1.2	1-Needle	L3*	**
	Hardness test	3.5	4.4.2	1-Needle	L1*	**
	Bend test	3.6	4.4.3	1-Needle	L2*	**

* A minimum of three units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one needle.

4.2.2. *Sampling for examination.* A random sample of needles shall be selected from each lot offered to the Government in accordance with Military Standard MIL–STD–105. The Acceptable Quality Levels (AQL's) shall be as indicated below.

Type defect	Inspection Level	AQL (percent defective)
Major defects	II	1.0
Minor A	II	2.5
Minor B	L3	4.0

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, and as indicated in table III.

4.2.4 *Raw materials.* A sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in table IV. Any needles in the sample containing one or more defects shall be considered as one defective needle and if the number of defective needles in any sample exceeds the acceptance number of that sample, the lot represented by the sample shall be rejected.

TABLE IV.—*Classification of defects in accordance with Military Standard MIL–STD–105*

Categories	Defects
Critical:	*None defined.*
Major:	
101.	Needle not of type, class, point, and size specified.
102.	Needle bent, blunt (except where specified) not free of feathered edges, burrs, filings, dirt, or foreign matter.
103.	Needle not free of cracks (particularly at eye sections) or unspecified bends.
104.	Eyehole of needle omitted in whole or in part, or not smooth.
Minor A:	
201.	Needles generally pitted or otherwise not smoothly or symmetrically finished.
202.	Suture grooves omitted in whole or in part.
Minor B:	
251.	Dimensions not within tolerances shown in applicable figure.

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4.4 Tests.

4.4.1 *Corrosion-resistance tests.*

4.4.1.1. *Boil test.* Sample needles shall be washed with soap and warm water, rinsed in distilled water, and dried. They shall then be boiled for 30 minutes in distilled water in a glass beaker and shall remain submerged for 48 hours, then allowed to dry and examined under 4 to 7 power magnification.

4.4.1.2 *Copper sulfate test.* Sample needles shall be washed with soap and warm water, rinsed in hot water, followed by a dip in ethyl alcohol (95 percent), and allowed to dry. They shall then be completely immersed in the copper sulfate solution at a temperature of 65° ± 2° Fahrenheit (F.) for 6 minutes and shall then be washed off with a cloth saturated with fresh water. The copper sulfate solution shall be made up as follows:

Copper sulphate ($Cu\ SO_4\ .5\ H_2O$) .... 4.0 grams
Sulphuric acid ($H_2\ SO_4$) (Specific gravity 1.84) .....10.0 grams
Water (H_2O) .....90.0 milliliters

4.4.2 *Hardness teast.* Hardness shall be determined in accordance with the Rockwell or Diamond Pyramid methods as described in methods 243 and 244 of Federal Test Method Standard No. 151.

4.4.3 *Bend test.* The needles shall be tested as described in method 231 of Federal Test Method Standard No. 151 for longitudinal bend test. The needles shall be bent through an angle of 45° over a die, or round edge, of a diameter equal to the wire size of the needle tested.

4.4.4 *Determination of composition.* Composition shall be determined as specified in method 111 or 112 of Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

For civil agency procurements the definitions and applications of the various levels of packaging and packing shall be as specified in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A or B.*

5.1.1.1. *Types I through X, types XII through XVI, and type XVIII.*

5.1.1.1.1 *Unit package.* Needles, in quantities specified in table V, shall be completely wrapped or packaged in an envelope. Wrapping or envelope shall be fabricated from transparent, anticorrosive, plastic material. The envelope-type package shall be further packaged in a transparent plastic holder, open at one end, with thumb notch. The wrapped-type package shall be further packaged in a transparent plastic bag heat sealed on all edges.

TABLE V. *Packaging quantities*

Types	Number of needles per unit of package	Number of unit packages per intermediate package	Number of intermediate packages per exterior container
I through X, XII through XVI, and XVIII	6 needles	24	60
XI and XVII	1 needle	24	48

5.1.1.1.2 *Alternate unit package.* As an alternate to the unit package specified above, 6 needles may be packaged in an envelope or folder of suitable size and design, fabricated of anticorrosive material. Envelope or folder shall be adequately secured to prevent accidental opening. Each container shall be further packaged in accordance with method IA-8 of Military Specification MIL-P-116.

5.1.1.2 *Types XI and XVII.*

5.1.1.2.1 *Unit package.* Each needle shall have its working end adequately cushioned to protect it from damage and to prevent it from

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penetrating the wrapper. Each needle so protected shall be wrapped in tissue paper and packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening.

5.1.1.3 *Intermediate package.* Unit packages shall be intermediately packaged in quantities as specified in table V in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.2 *Level C.* Needles shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* Needles, packaged as specified in 5.1.1, shall be packed in quantities as specified in table V in exterior containers constructed in accordance with Federal Specification PPP–B–621, table III. Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.* Needles, packaged as specified in 5.1.1, shall be packed in quantities as specified in table V in exterior containers of appropriate size constructed in accordance with Federal Specification PPP–B–621, table I, or PPP–B–636, type I, class 1, and fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, unit, intermediate, and shipping containers shall be marked in accordance with Federal Standard No. 123. For transparent packages, the required information shall be printed on label and inserted within the packages.

5.3.2 *Military activities.*

5.3.2.1 Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor. For transparent package, the required information shall be printed on a label inserted within the package in such manner as to permit ready identification. Barrier bag for alternate package shall bear the same information and in addition, the legend: "Package Method IA–8X".

5.3.2.2 *Intermediate and shipping container.* Each intermediate and shipping container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, point, and size required (see 1.2).

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specifica-

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tion, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Needles, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

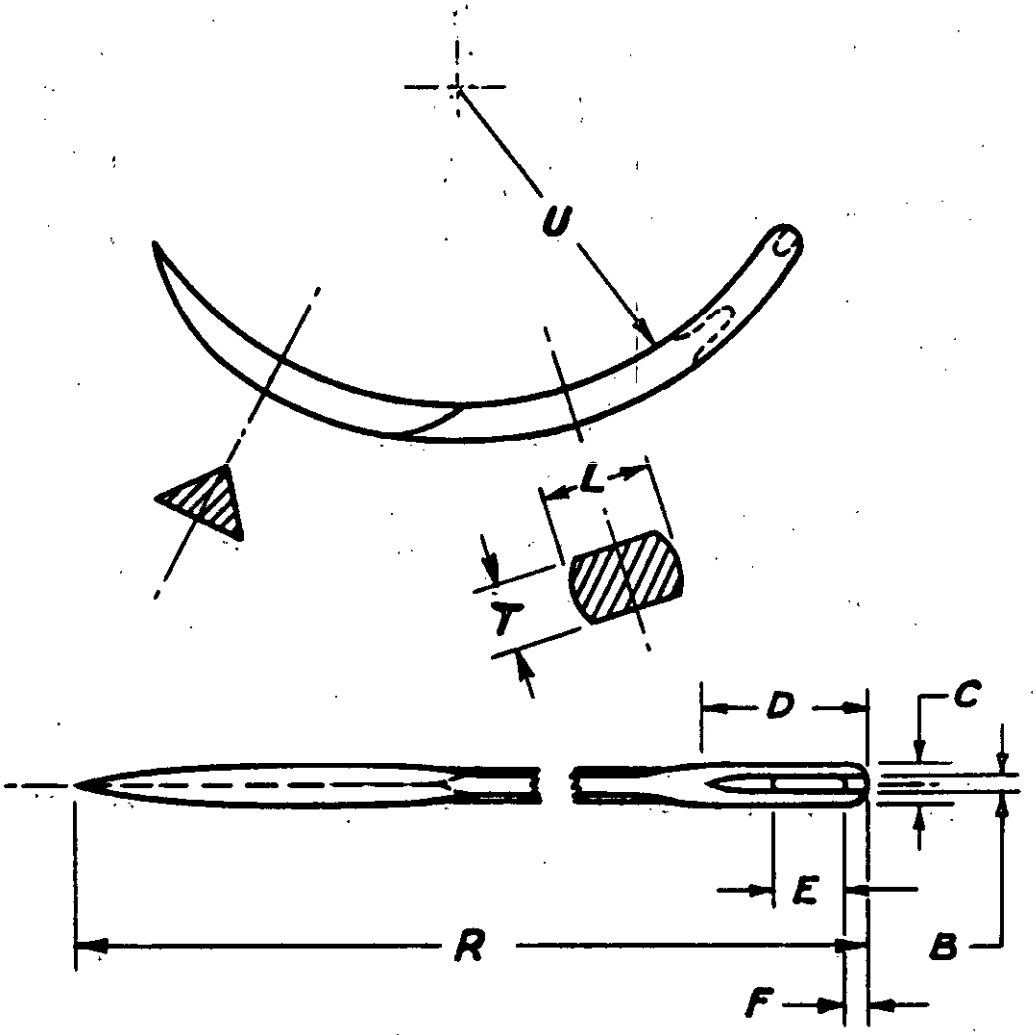
Needles, not otherwise indexed.
Motor volume minimum weight 20,000 pounds.

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MILITARY CUSTODIANS:

Army—M
Navy—Md
Air Force—AFCSC.

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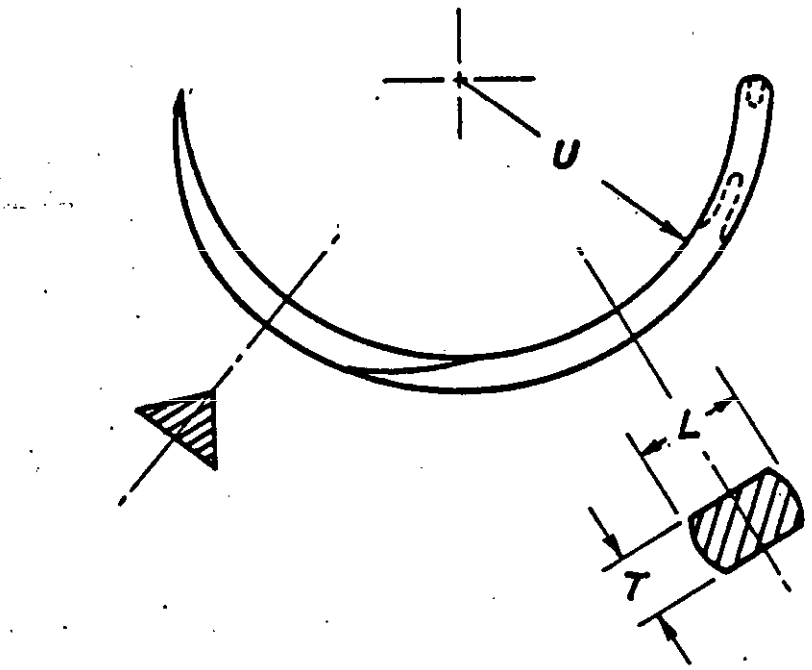
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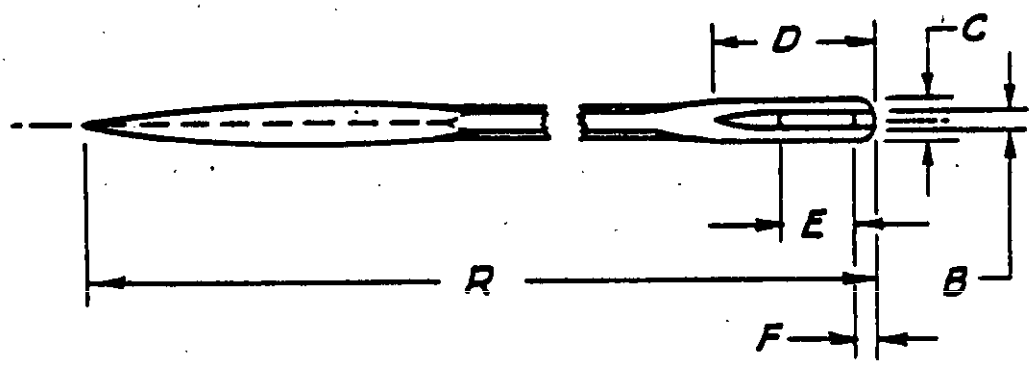
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4	.046	.040	.070	11/32	.100	.046	.050	2-5/8	.040	1-1/16
6	.046	.033	.062	11/32	.100	.046	.050	2-7/16	.040	31/32
8	.040	.030	.055	21/64	.080	.040	.044	2-3/16	.032	7/8
10	.035	.025	.050	9/32	.070	.035	.038	1-7/8	.026	3/4
12	.030	.024	.046	3/16	.0625	.030	.036	1-9/16	.024	5/8
14	.028	.022	.042	3/16	.0625	.028	.032	1-5/16	.024	17/32
16	.028	.022	.042	3/16	.0625	.028	.032	1-3/16	.024	15/32
18	.025	.018	.036	3/16	.050	.025	.027	1-3/32	.020	3/8
20	.025	.018	.036	3/16	.050	.025	.027	7/8	.020	5/16

A = WIRE SIZE

FIGURE 1—Type I, Class A, Point 1

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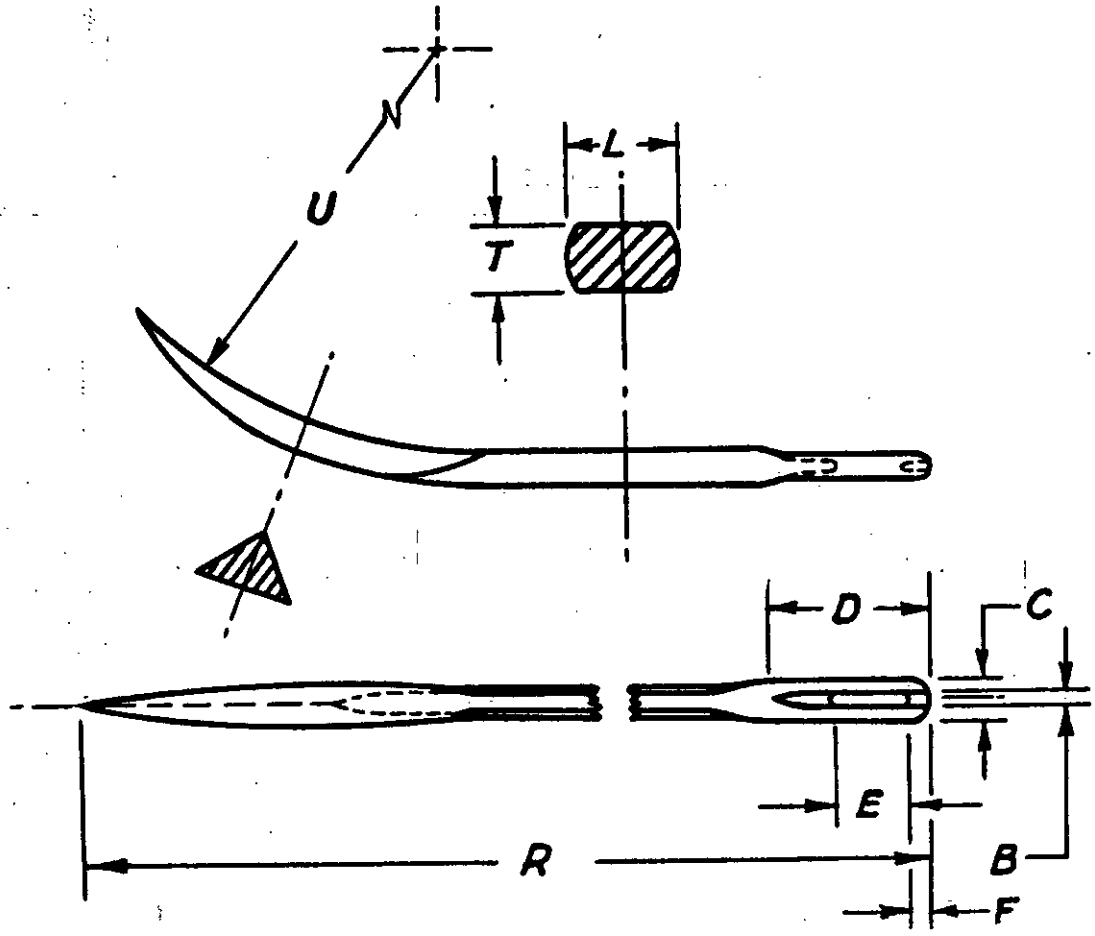
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Size	A	B	C	D	E	F	L	R	T	U
1	.046	.033	.062	11/32	.100	.046	.050	2-1/16	.040	43/64
2	.040	.030	.055	21/64	.080	.040	.044	1-31/32	.032	5/8
3	.040	.030	.055	21/64	.080	.040	.044	1-49/64	.032	9/16
4	.035	.025	.050	9/32	.070	.035	.038	1-43/64	.026	1/2
5	.035	.025	.050	9/32	.070	.035	.038	1-3/8	.026	7/16

A = WIRE SIZE

FIGURE 2—Type I, Class B, Point 1

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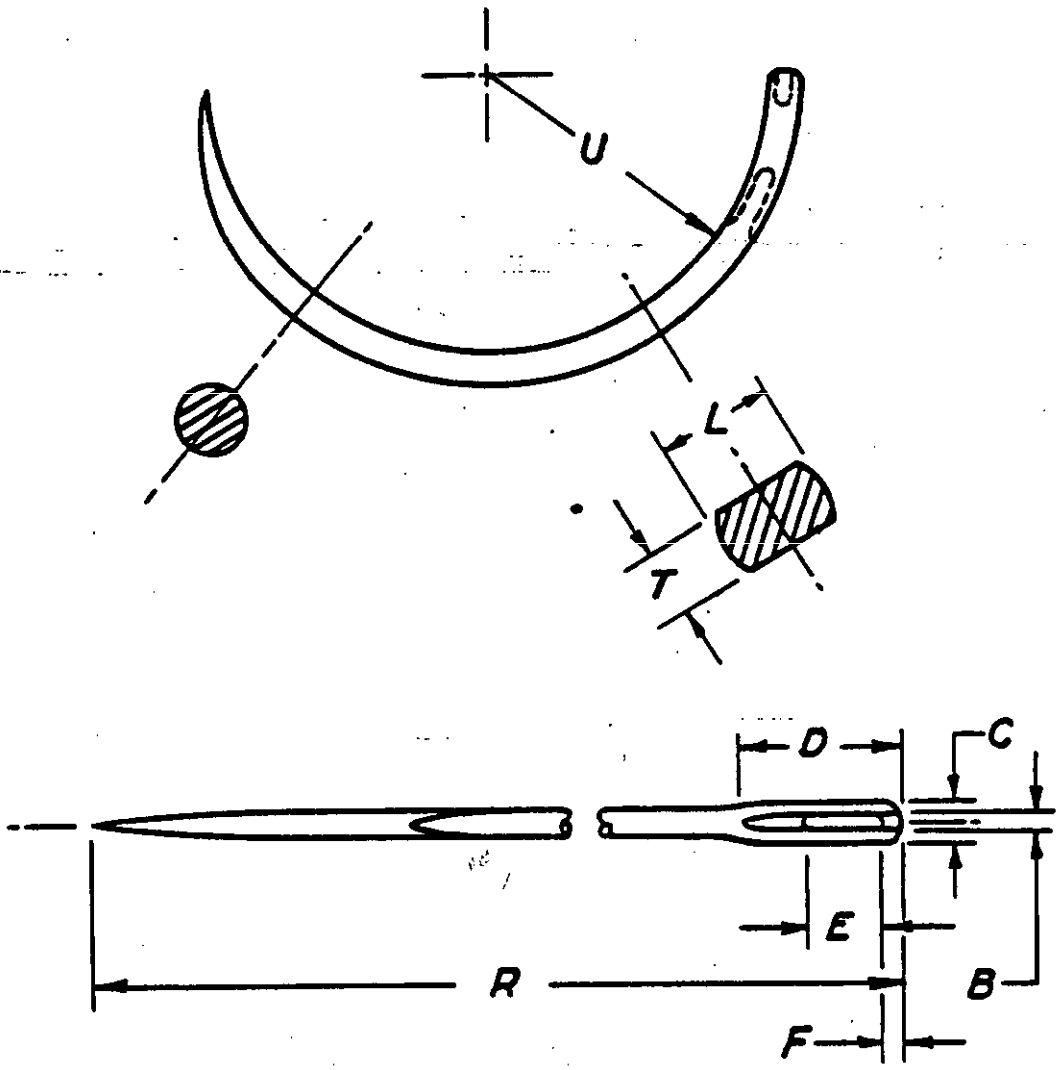
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Size	A	B	C	D	E	F	L	R	T	U
10	.040	.030	.055	21/64	.080	.040	.044	1-45/64	.032	15/16
12	.035	.025	.050	9/32	.070	.035	.038	1-29/64	.026	13/16
14	.030	.024	.046	3/16	.0625	.030	.036	1-9/32	.024	21/32
16	.030	.024	.046	3/16	.0625	.030	.036	1-7/64	.024	19/32
18	.025	.018	.036	3/16	.050	.025	.027	29/32	.020	15/32

A = WIRE SIZE

FIGURE 3—Type I, Class C, Point 1

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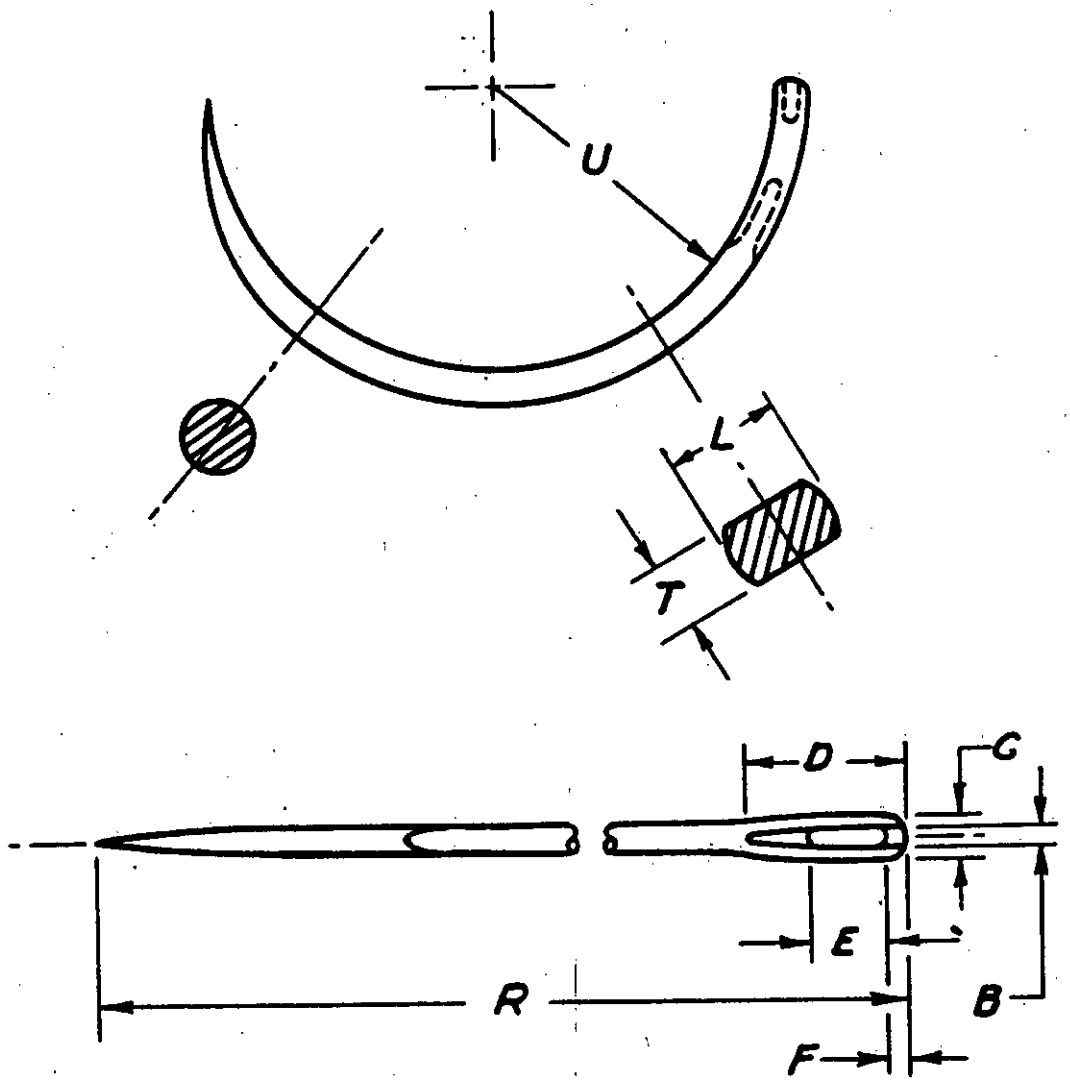
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Size	A	B	C	D	E	F	L	R	T	U
1	.025	.018	.036	3/16	.050	.025	.027	2-1/4	.020	11/16
2	.025	.018	.036	3/16	.050	.025	.027	2	.020	5/8
3	.025	.018	.036	3/16	.050	.025	.027	1-3/4	.020	9/16
4	.025	.018	.036	3/16	.050	.025	.027	1-1/2	.020	15/32
5	.023	.016	.034	9/32	.050	.025	.025	1-3/8	.018	7/16
6	.023	.016	.034	9/32	.050	.025	.025	1-1/4	.018	3/8

A = WIRE SIZE

FIGURE 4—Type II, Class B, Point 3

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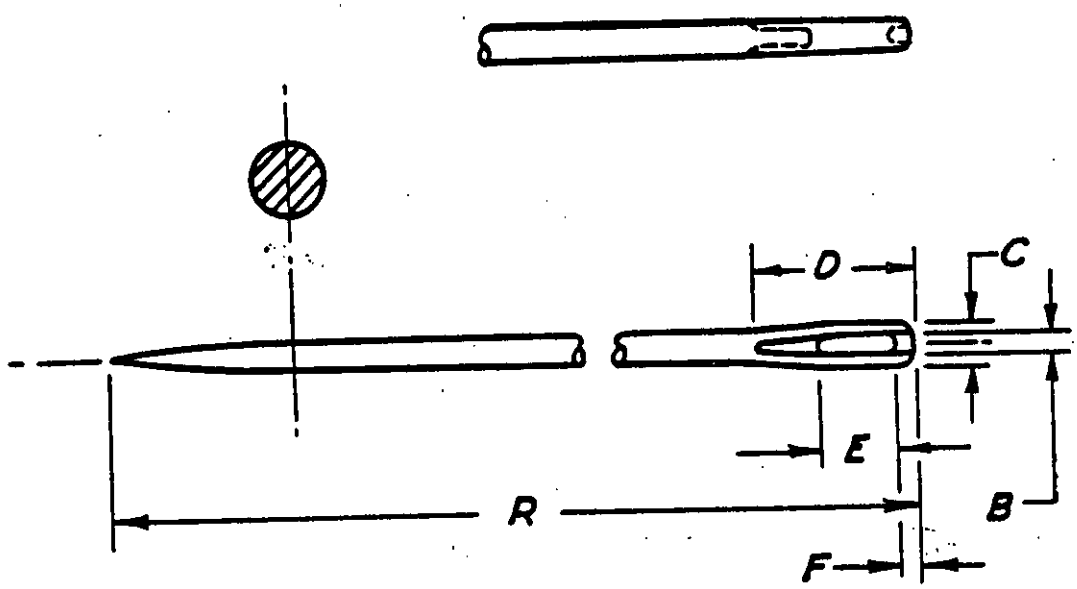
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Size	A	B	C	D	E	F	L	R	T	U
2	.040	.030	.055	21/64	.080	.040	.044	1-11/16	.032	33/64
3	.035	.025	.050	9/32	.070	.035	.038	1-7/16	.026	29/64
4	.035	.025	.050	9/32	.070	.035	.038	1-1/4	.026	25/64

A = WIRE SIZE

FIGURE 5—Type III, Class B, Point 3

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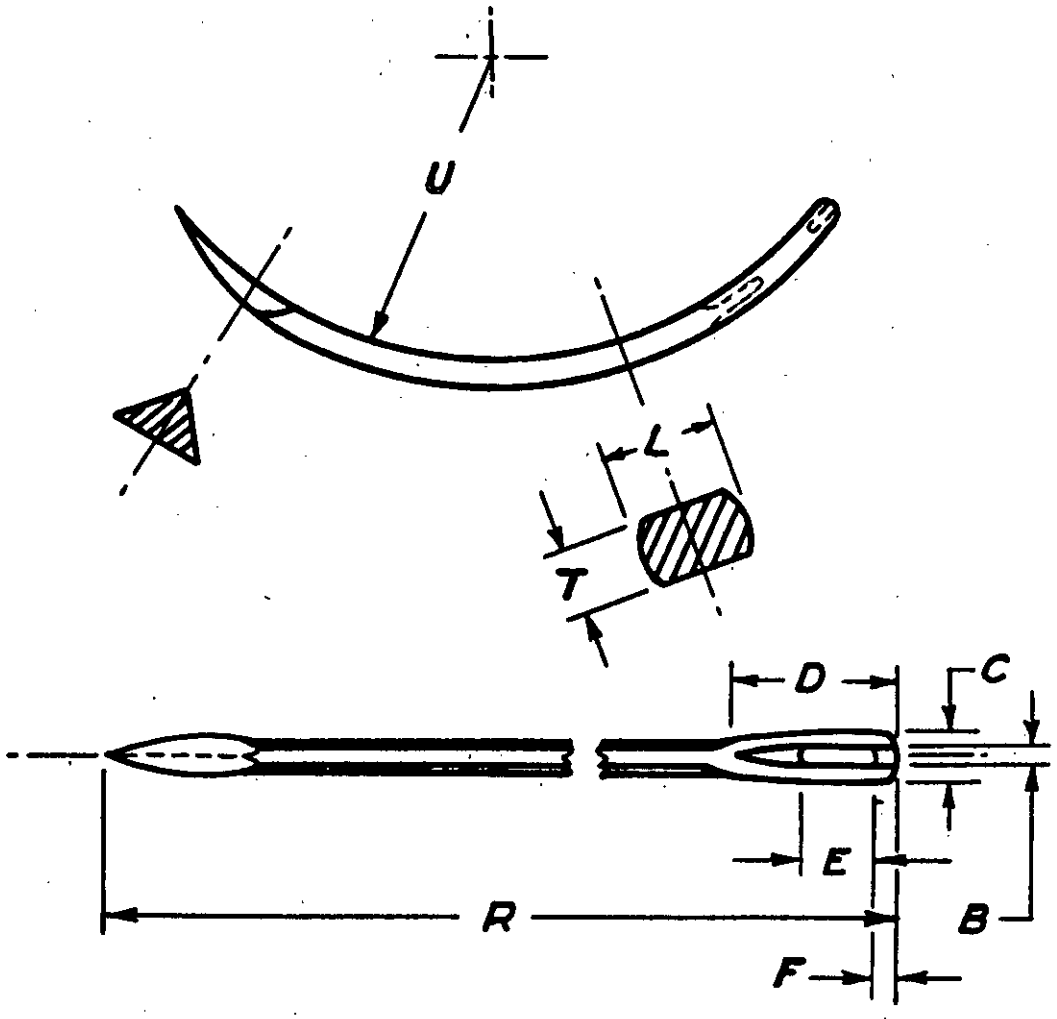


Size	A	B	C	D	E	F	R
2-3/4	.030	.024	.046	3/16	.0625	.030	2-3/4
2-1/2	.030	.024	.046	3/16	.0625	.030	2-1/2
2	.030	.024	.046	3/16	.0625	.030	2
1-3/4	.028	.022	.042	3/16	.0625	.028	1-3/4
1-1/2	.025	.018	.036	3/16	.050	.025	1-1/2

A = WIRE SIZE

FIGURE 6—Type IV, Class D, Point 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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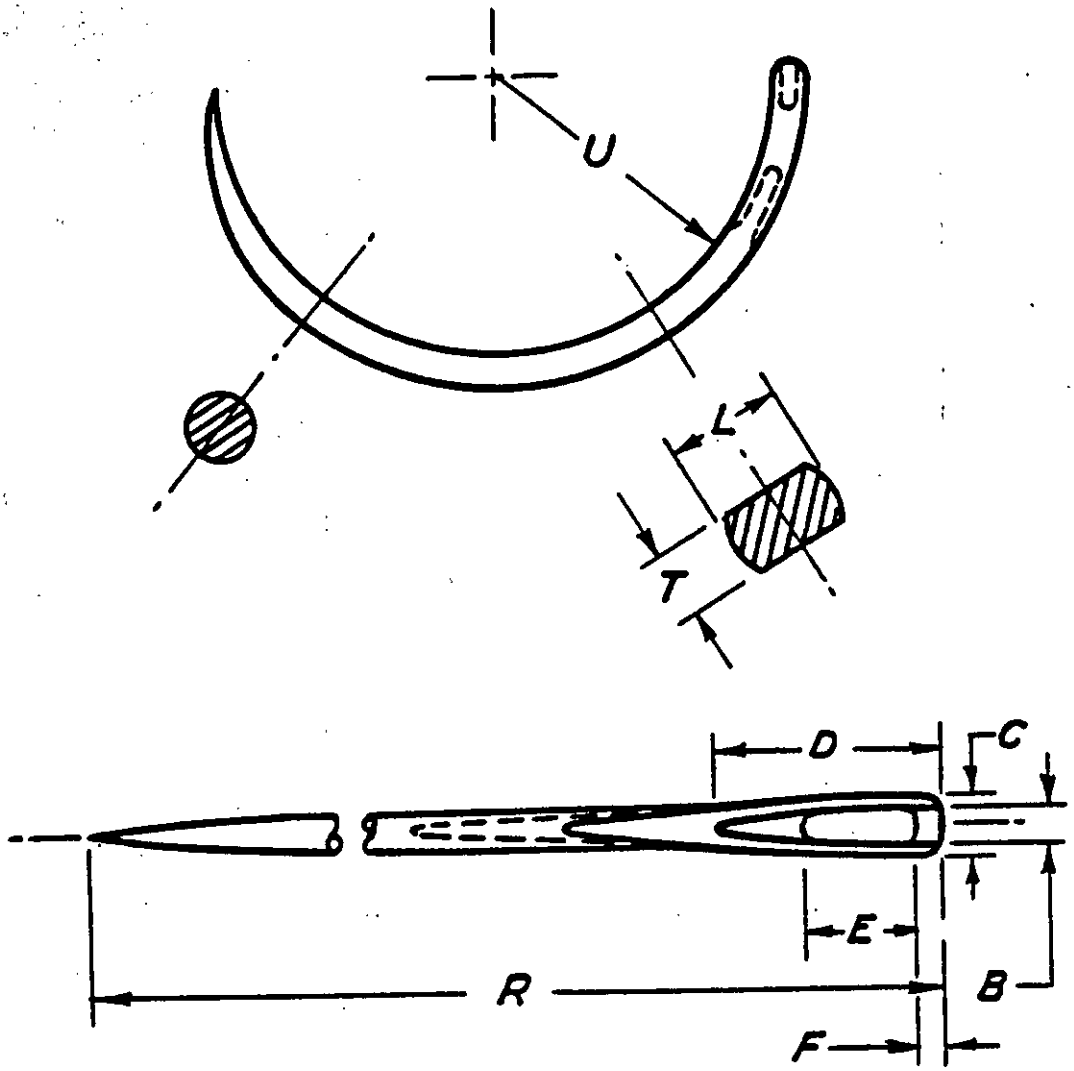
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.048	.036	.075	11/32	.100	.048	.053	2-1/16	.042	1-5/16
3	.046	.036	.075	11/32	.100	.048	.053	1-1/2	.042	21/32

A = WIRE SIZE

FIGURE 7—Type V, Class A, Point 2

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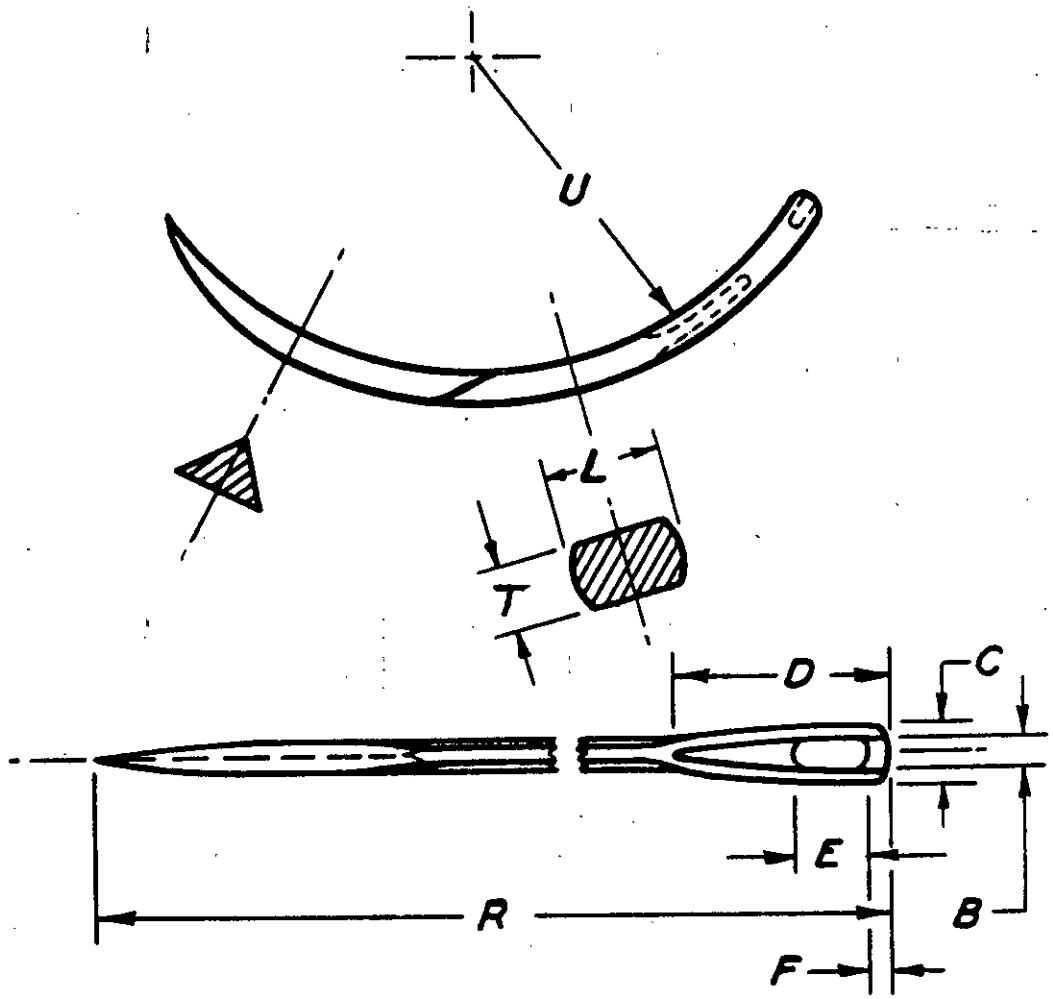
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.050	.038	.075	11/32	.110	.050	.052	2-1/16	.044	5/8
2	.050	.038	.075	11/32	.110	.050	.052	1-13/16	.044	9/16
3	.050	.038	.075	11/32	.110	.050	.052	1-5/8	.044	1/2
4	.048	.036	.075	11/32	.100	.048	.053	1-3/8	.044	13/32
6	.039	.030	.054	15/64	.080	.039	.043	1-1/64	.032	11/32

A = WIRE SIZE

FIGURE 8—Type V, Class B, Point 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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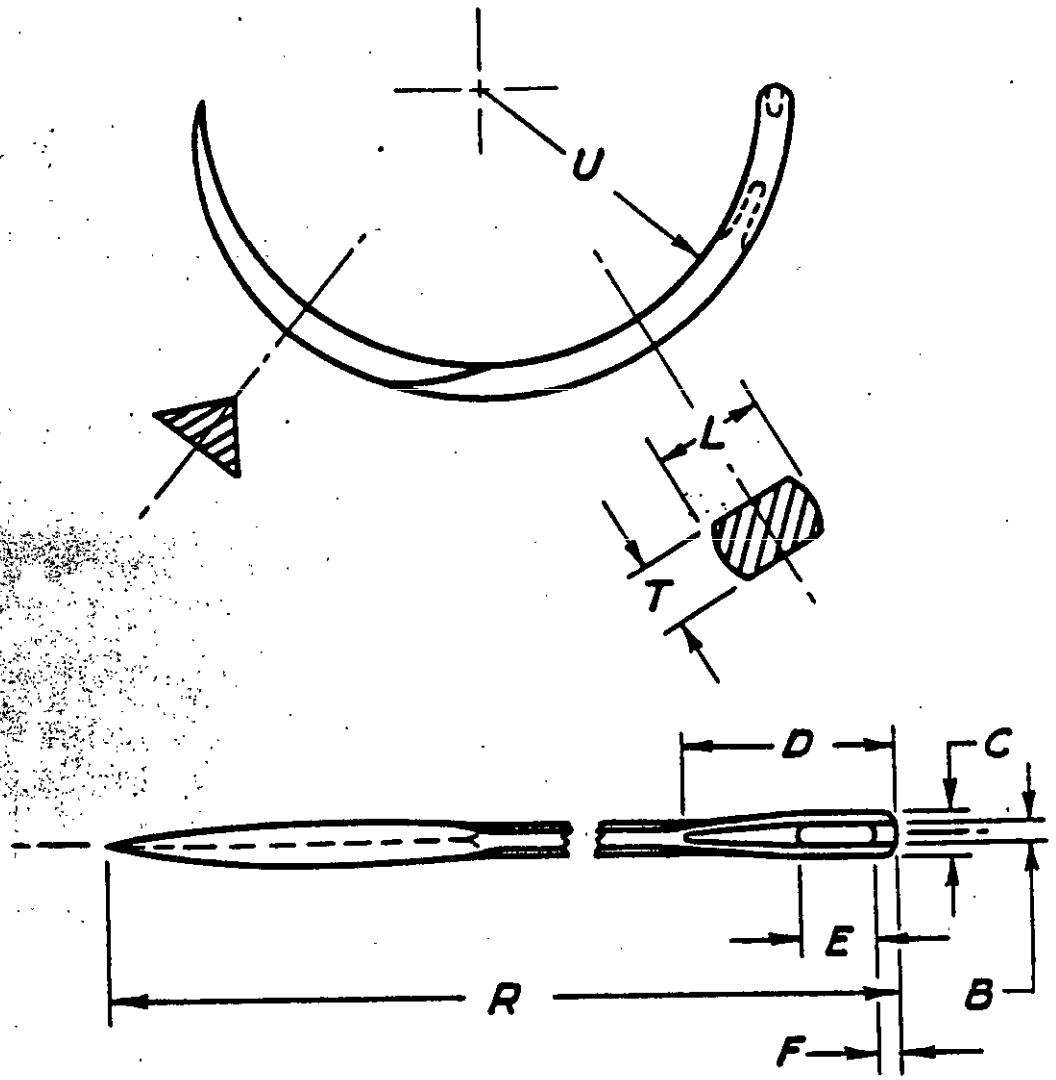
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.016	.013	.025	1/8	.045	.020	.021	1-1/8	.011	7/16
2	.016	.013	.025	1/8	.045	.020	.021	1	.011	7/16
3	.016	.013	.025	1/8	.045	.020	.021	7/8	.011	23/64
4	.016	.013	.025	1/8	.045	.020	.021	5/8	.011	1/4

A = WIRE SIZE

FIGURE 9—Type VI, Class A, Point 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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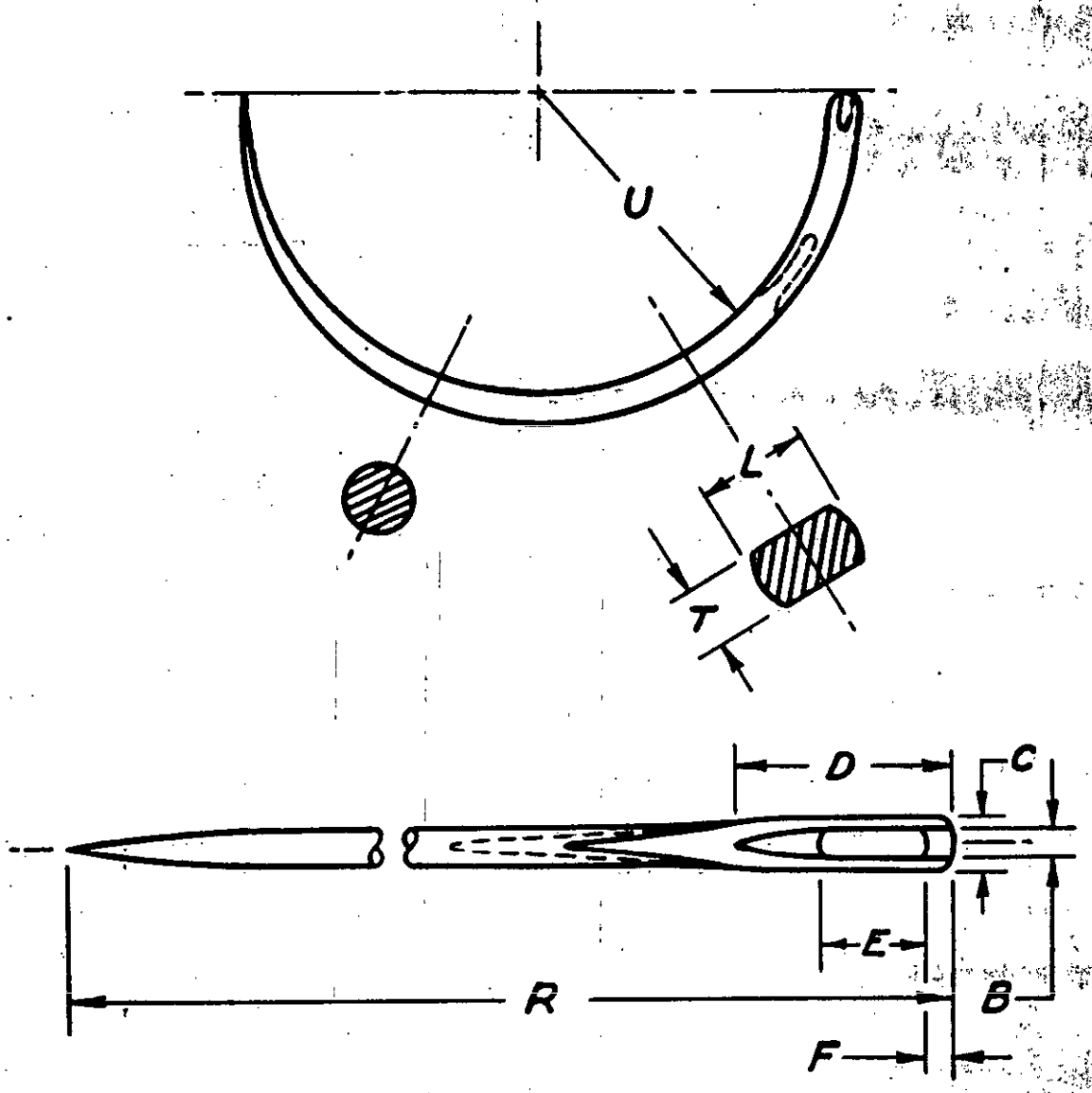
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.016	.013	.025	1/8	.045	.020	.021	1-1/8	.011	11/32
2	.016	.013	.025	1/8	.045	.020	.021	1	.011	5/16
3	.016	.013	.025	1/8	.045	.020	.021	7/8	.011	9/32
4	.016	.013	.025	1/8	.045	.020	.021	5/8	.011	3/16

A = WIRE SIZE

FIGURE 10—Type VI, Class B, Point 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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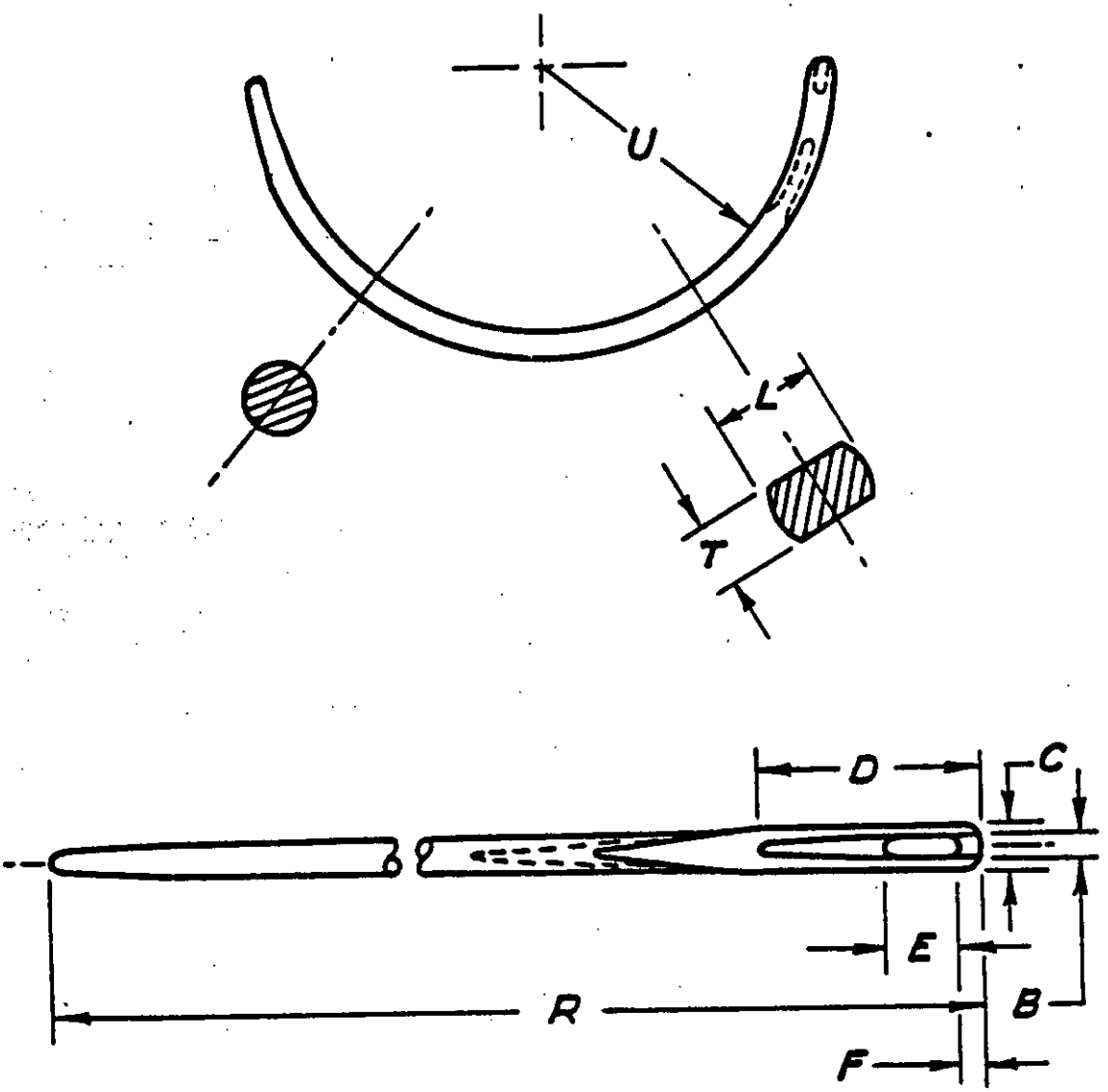
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.016	.013	.025	1/8	.045	.020	.021	1-1/8	.011	11/32
2	.016	.013	.025	1/8	.045	.020	.021	1	.011	5/16
3	.016	.013	.025	1/8	.045	.020	.021	7/8	.011	9/32
4	.016	.013	.025	1/8	.045	.020	.021	5/8	.011	3/16

A = WIRE SIZE

FIGURE 11—Type VI, Class B, Point 3

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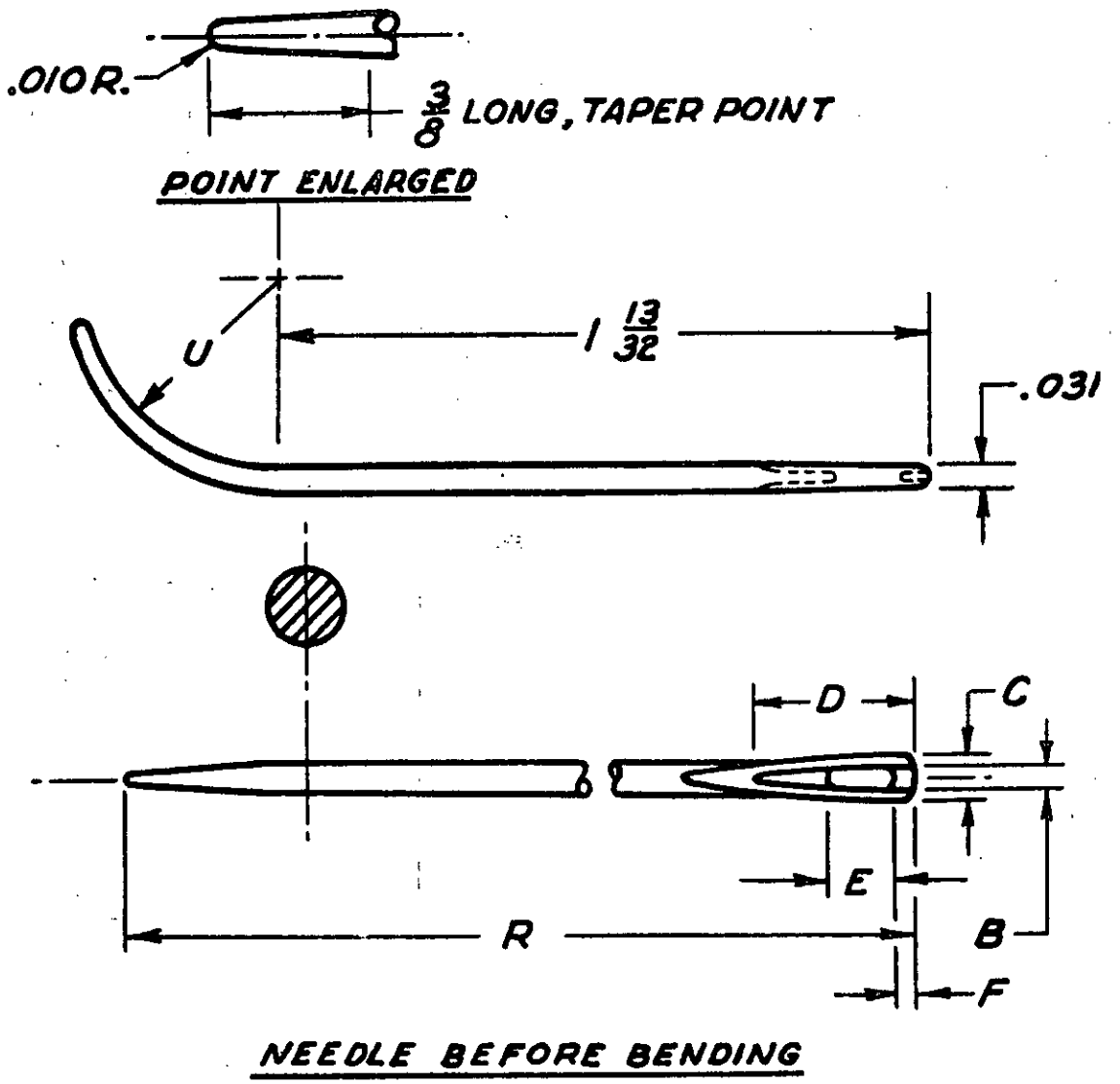
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
2-5/3	.050	.033	.075	11/32	.110	.050	.052	2-5/8	.044	7/8

A = WIRE SIZE

FIGURE 12—Type VII, Class B, Point 4

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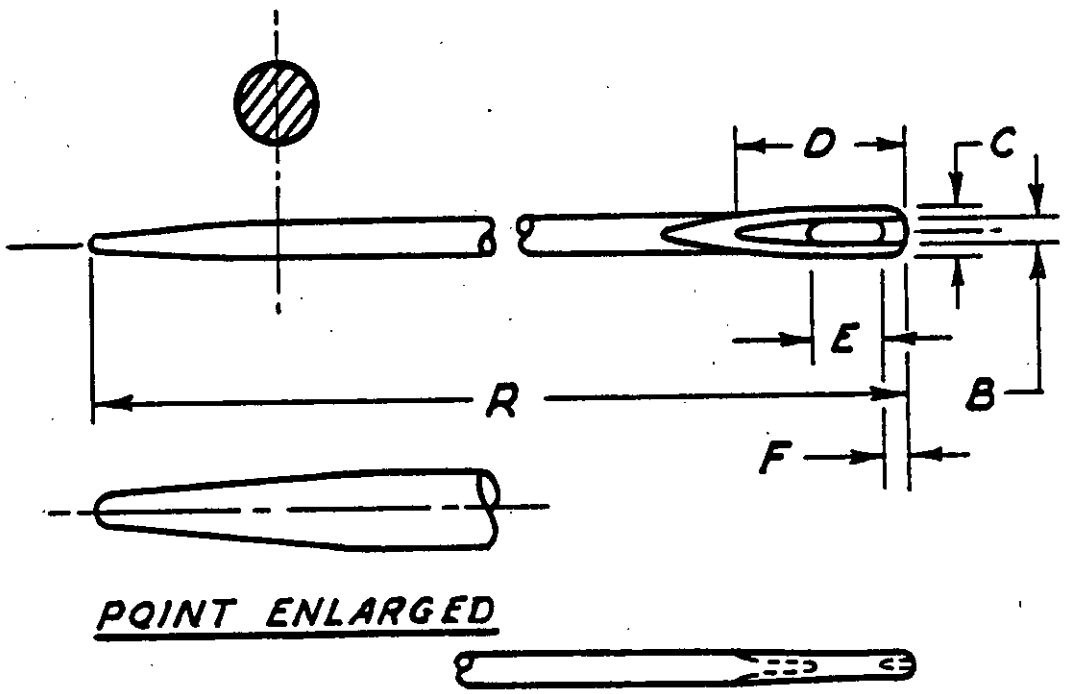


Size	A	B	C	D	E	F	R	U
2	.046	.033	.062	11/32	.100	.046	2	5/8

A = WIRE SIZE

FIGURE 13—Type VII, Class C, Point 4

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Check the source to verify that this is the current version before use.

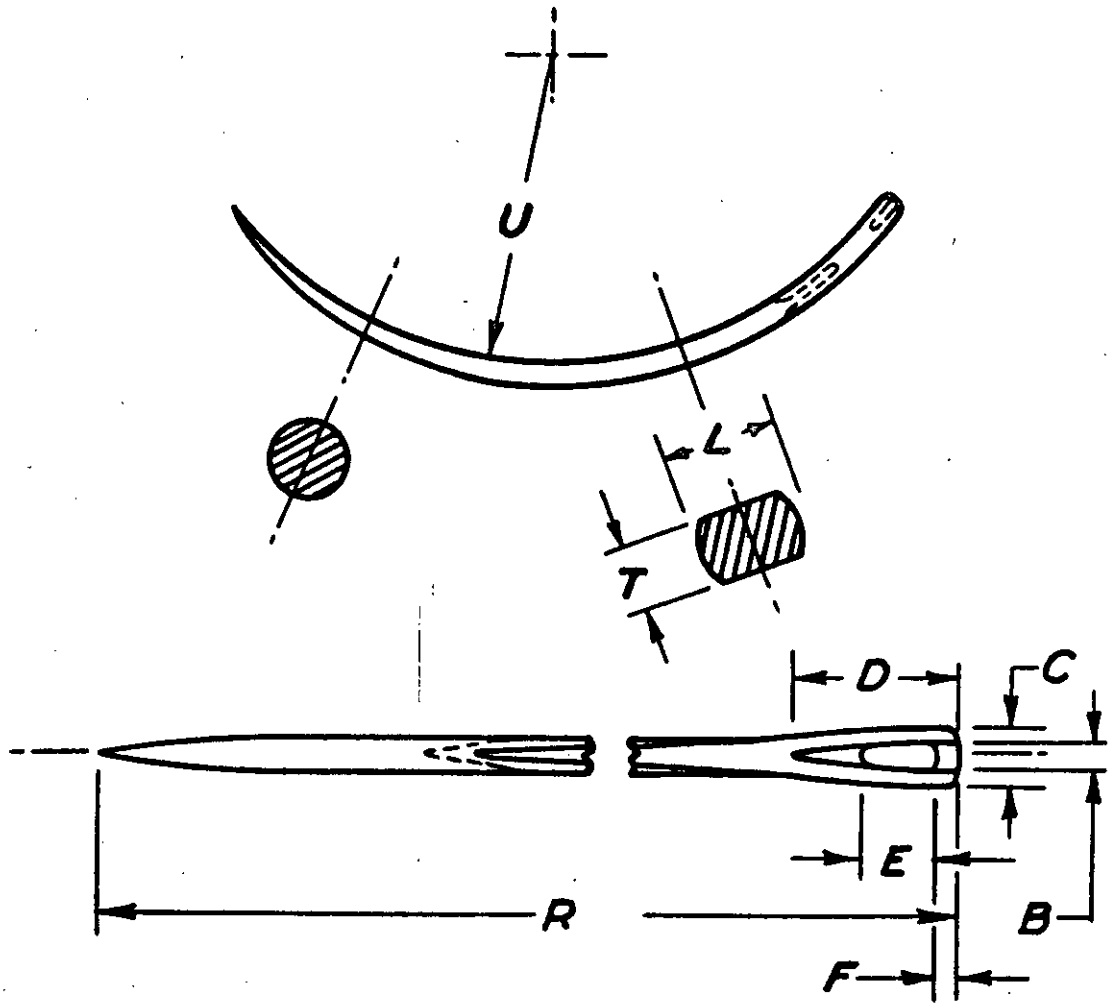


Size	A	B	C	D	E	F	R
1-3/4	.046	.033	.062	11/32	.100	.046	1-3/4

A = WIRE SIZE

FIGURE 14—Type VII, Class D, Point 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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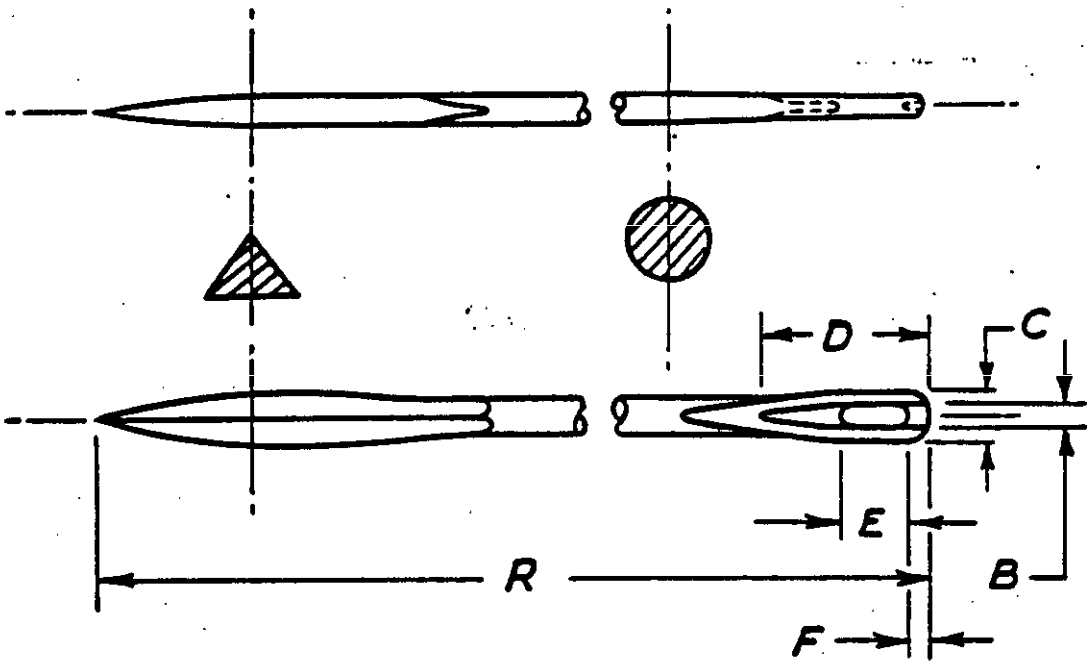
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
1	.054	.038	.075	25/64	.115	.054	.058	1-15/16	.048	13/16
2	.054	.038	.075	25/64	.115	.054	.058	1-11/16	.048	23/32
3	.054	.038	.075	25/64	.115	.054	.058	1-9/16	.048	21/32
4	.050	.038	.075	11/32	.110	.050	.052	1-1/4	.044	17/32
5	.046	.033	.062	11/32	.100	.046	.050	1-1/16	.040	29/64

A = WIRE SIZE

FIGURE 15—Type VIII, Class A, Point 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:49Z
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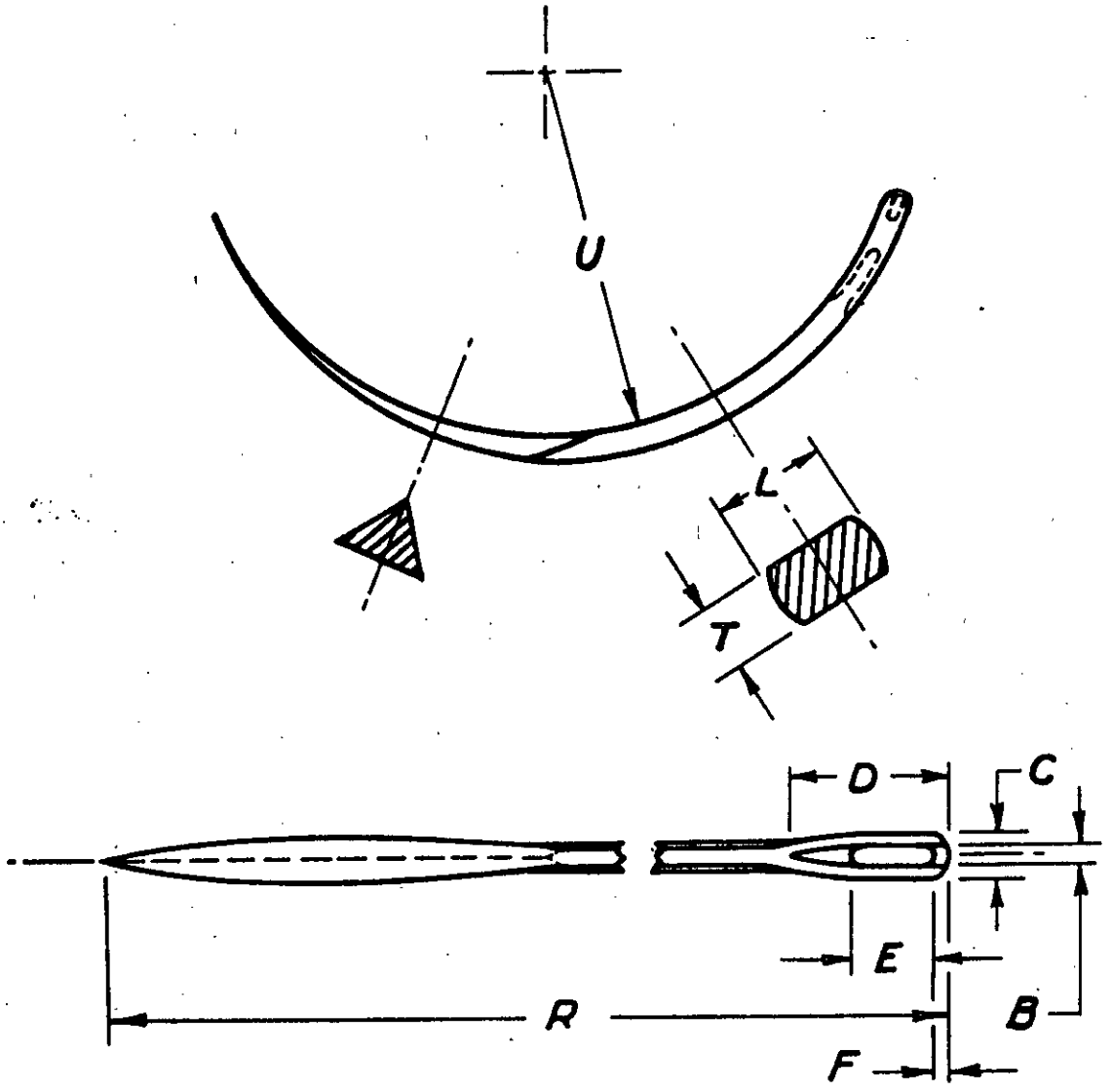


Size	A	B	C	D	E	F	R
3-1/2	.040	.025	.050	9/32	.070	.035	3-1/2
2-7/8	.035	.025	.050	9/32	.070	.035	2-7/8
2-1/2	.035	.025	.050	9/32	.070	.035	2-1/2
2	.035	.025	.050	9/32	.070	.035	2
1-3/4	.035	.025	.050	9/32	.070	.035	1-3/4
1-1/2	.035	.025	.050	9/32	.070	.035	1-1/2

A = WIRE SIZE

FIGURE 16—Type IX, Class D, Point 5

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Check the source to verify that this is the current version before use.



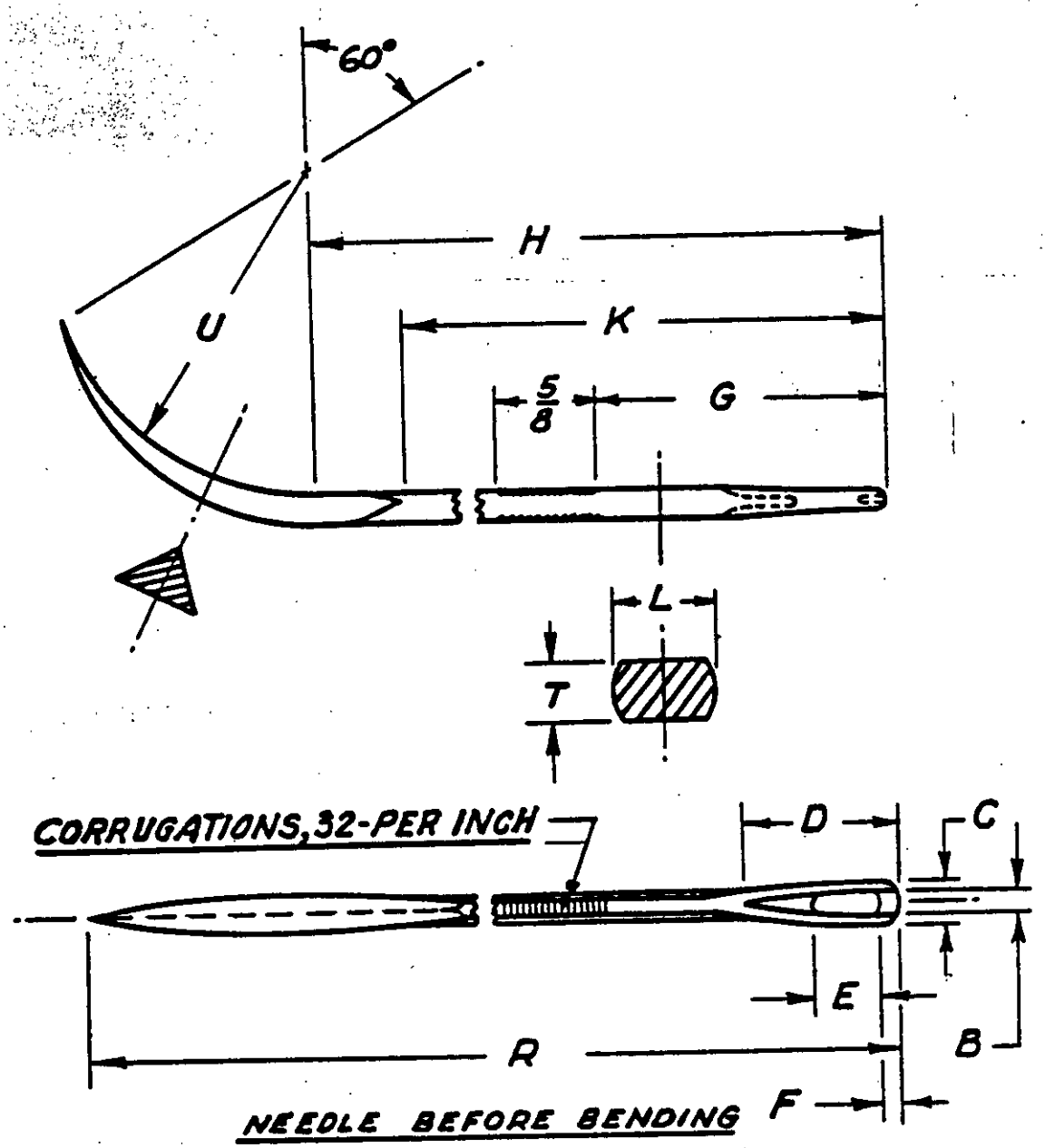
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
3-1/2 2-1/2	.046 .040	.033 .030	.062 .055	11/32 21/64	.100 .080	.046 .040	.050 .044	3-1/2 2-1/2	.040 .032	1-13/32 1

A = WIRE SIZE

FIGURE 17—Type X, Class A, Point 1

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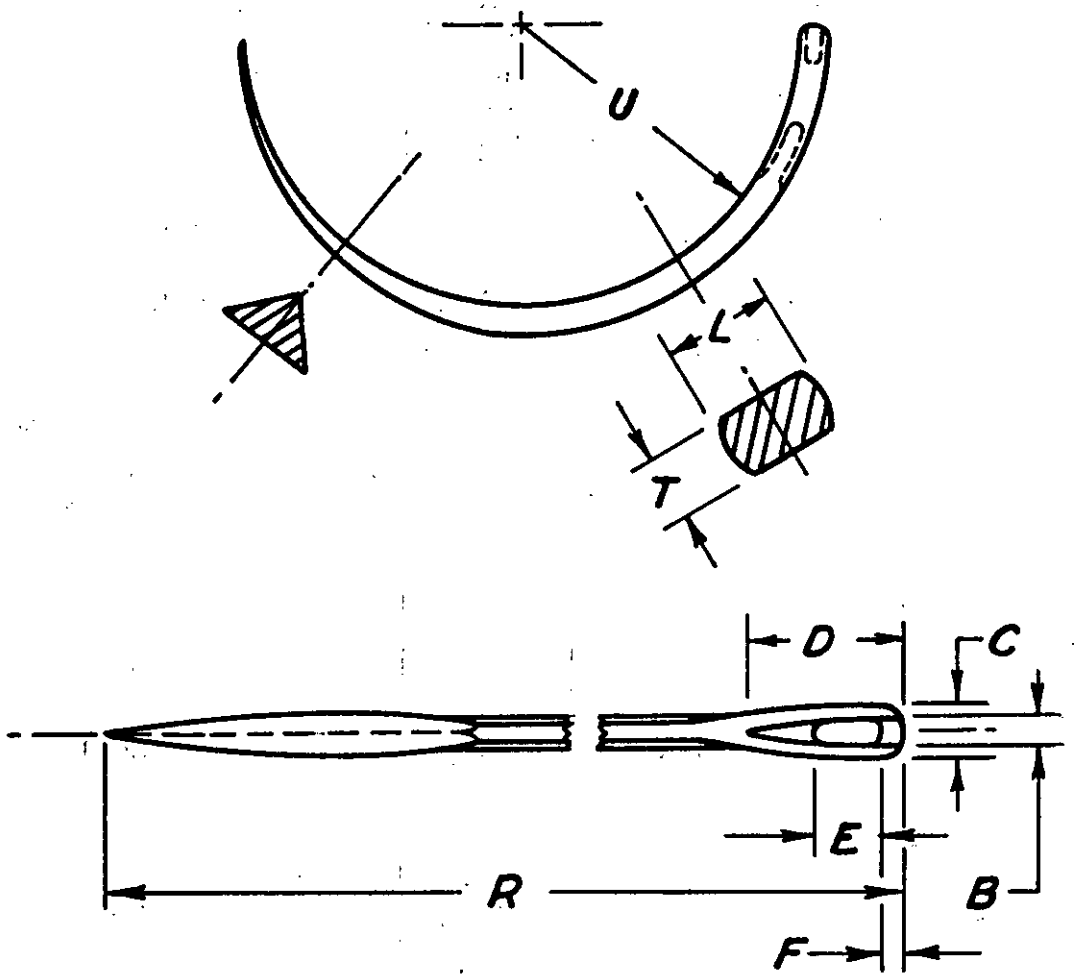


Size	A	B	C	D	E	F	G	H	K	L	R	T	U
3	.060	.040	.075	7/16	7/32	.060	11/16	1-5/8	1-1/2	.070	3	.050	1-3/8
5	.076	.065	.115	9/16	7/32	.075	1-1/4	3-1/8	2-3/4	.087	5	.066	1-3/4

A = WIRE SIZE

FIGURE 18—Type XI, Class C, Point 1

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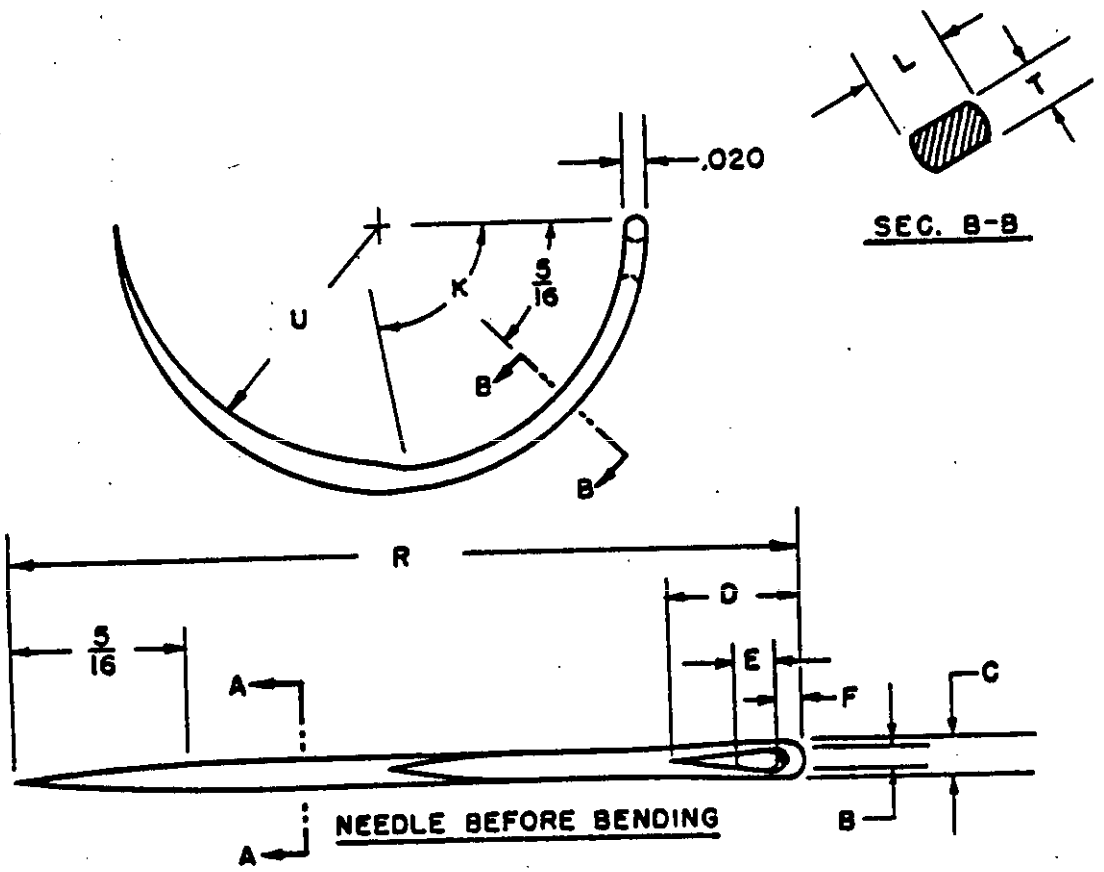
NEEDLE BEFORE BENDING

Size	A	B	C	D	E	F	L	R	T	U
0	.0625	.050	.093	13/32	.125	.077	.075	3-1/16	.045	7/8
1	.057	.050	.090	11/32	.115	.057	.072	2-5/8	.037	27/32
2	.054	.038	.075	11/32	.115	.054	.058	2-1/4	.044	23/32
3	.050	.038	.075	11/32	.110	.050	.054	2-1/16	.043	21/32
4	.048	.036	.075	11/32	.100	.048	.053	1-5/8	.039	1/2
5	.046	.033	.062	11/32	.100	.046	.050	1-7/16	.038	27/64
6	.040	.030	.055	11/32	.090	.042	.045	1-1/16	.031	11/32
7	.040	.030	.055	11/32	.090	.042	.045	15/16	.031	19/64

A = WIRE SIZE

FIGURE 19—Type XII, Class B, Point 1

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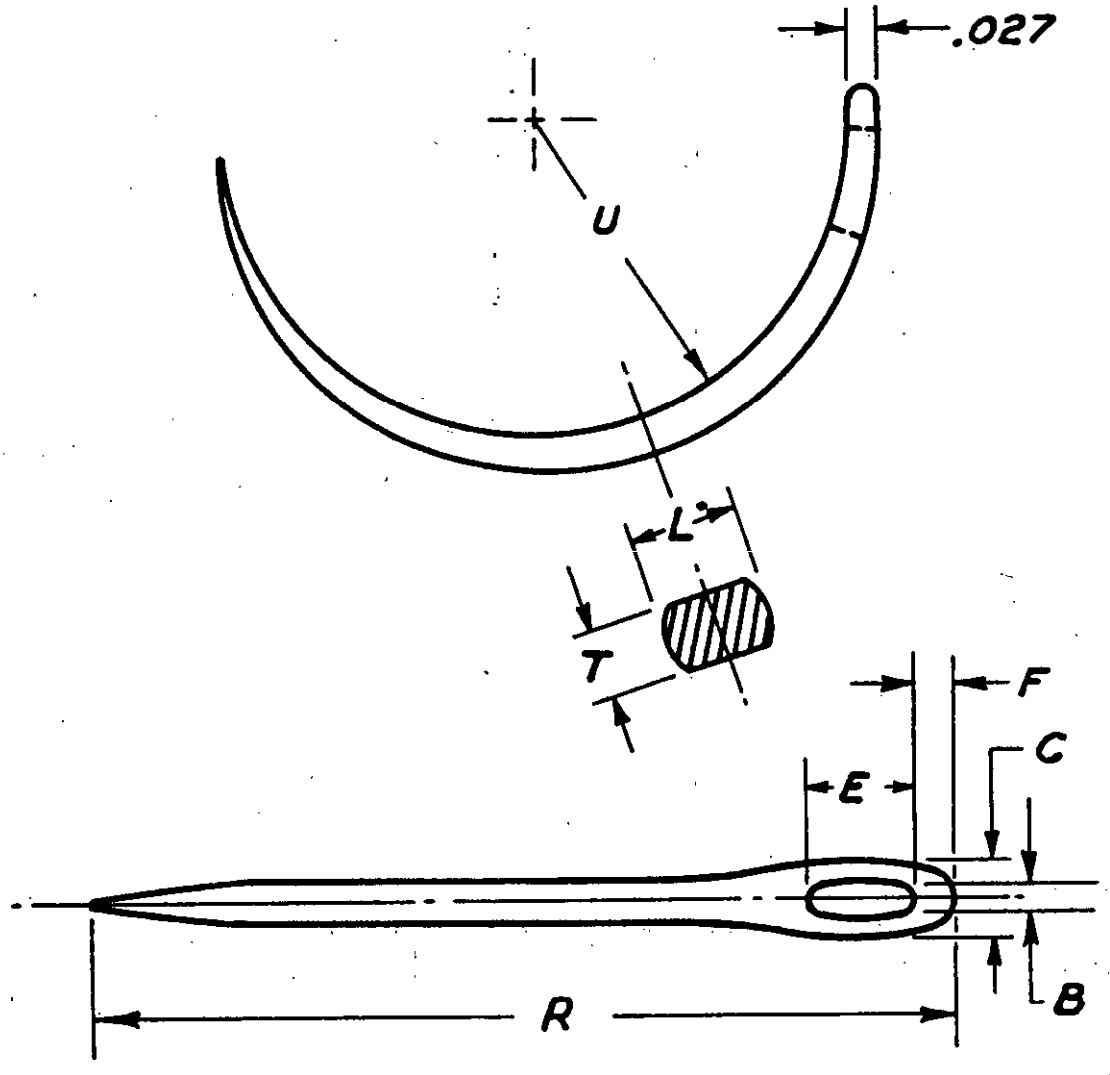


A	B	C	D	E	F	K	L	R	T	U
.030	.024	.046	5/16	.0625	.030	17/32	.036	1 1/16	.024	.338

A = WIRE SIZE

FIGURE 20—Type XIII, Class B, Point 3

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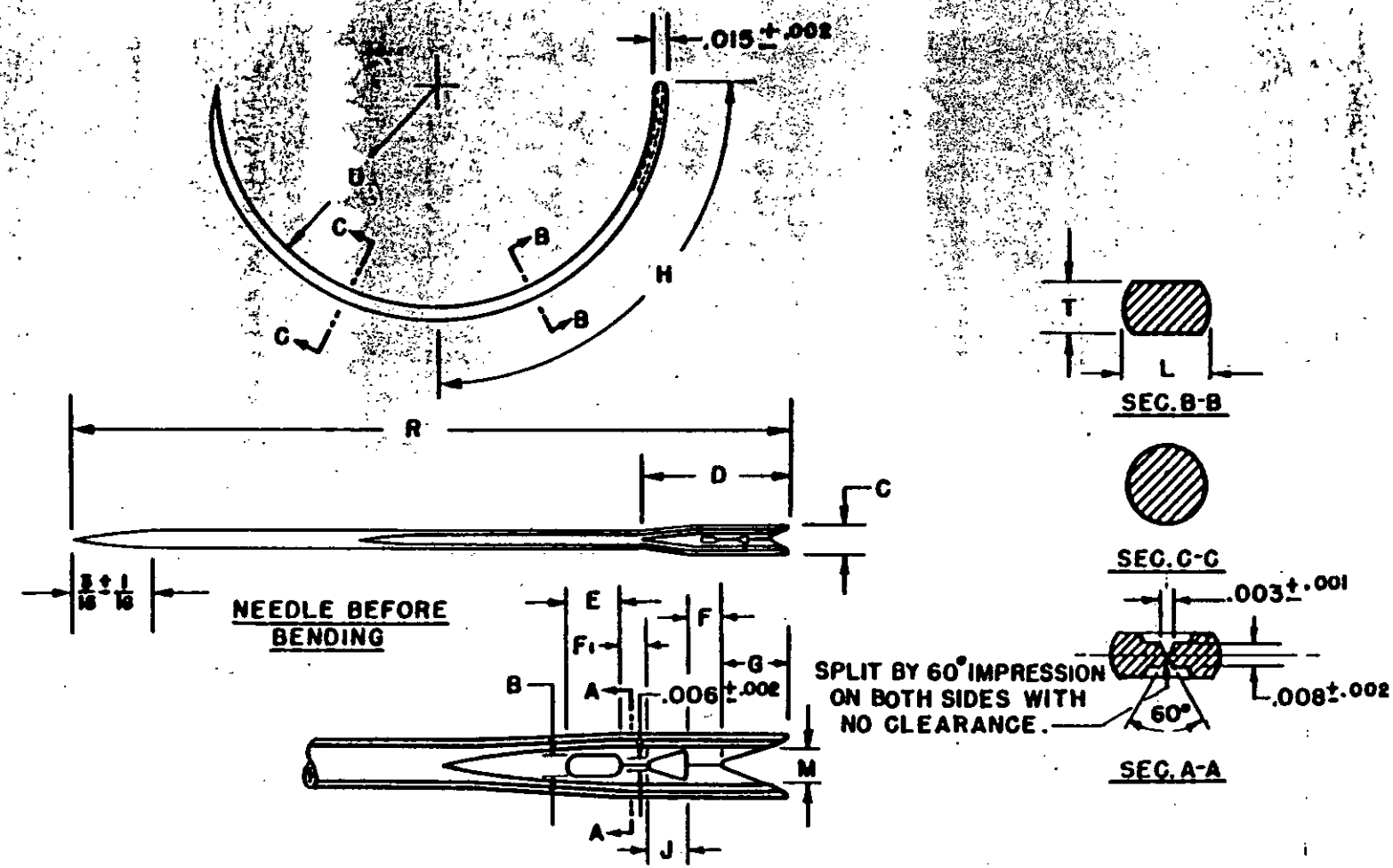
NEEDLE BEFORE BENDING

SIZE	A	B	C	E	F	L	R	T	U
1	.093	.092	.175	.226	.125	.118	1 1/2	.063	7/16
2	.093	.092	.175	.226	.125	.118	1 23/32	.063	35/64
3	.093	.092	.175	.226	.125	.118	2 3/8	.063	11/16

A = WIRE SIZE

FIGURE 21—Type XIV, Class B, Point 3

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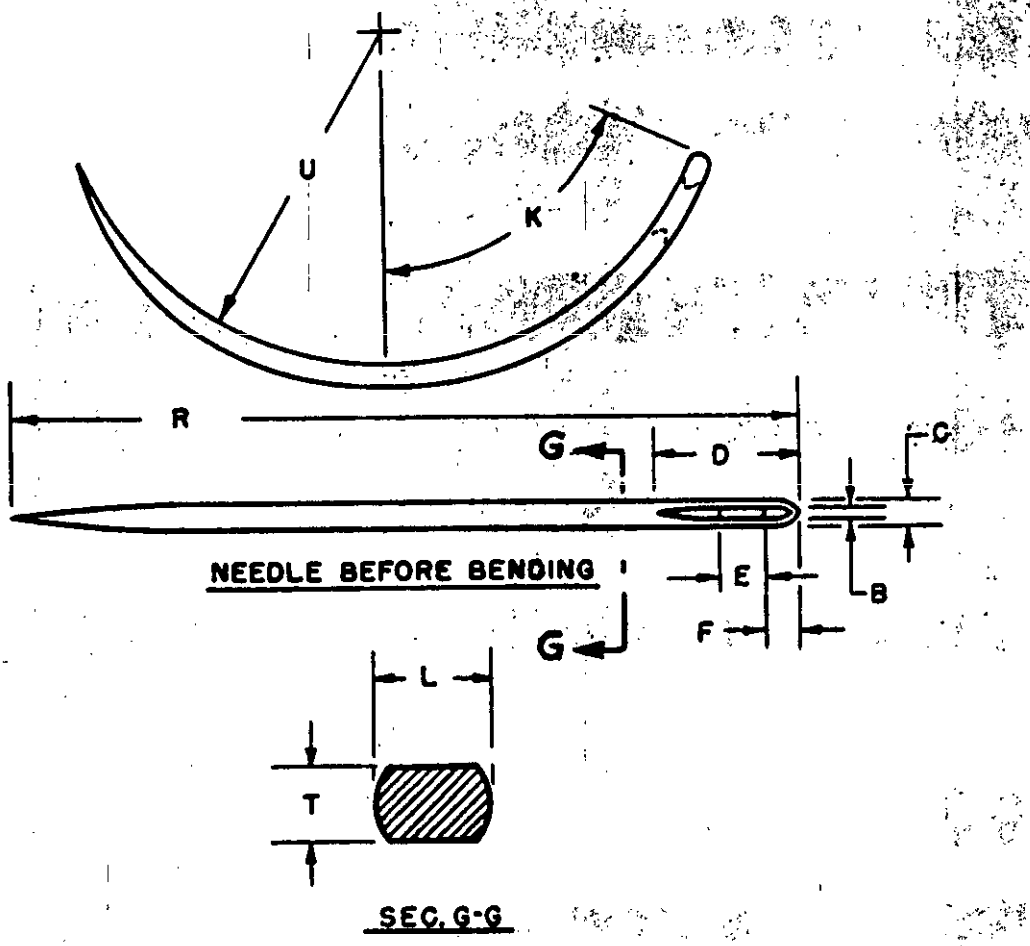


A	B	C	D	E	F1	F	G	H	J	L	M	R	T	U
.023	.012	.036	7/32	.035	.016	.019	.031	9/16	.031	.028	.018	1 1/8	.017	11/32

A= WIRE SIZE

FIGURE 22—Type XV, Class B, Point 3

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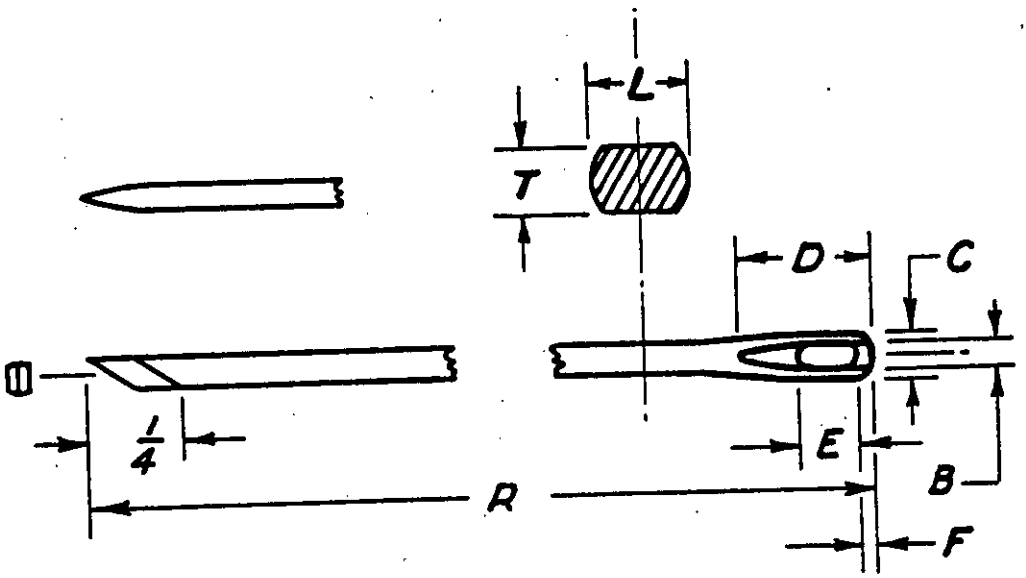


SIZE	A	B	C	D	E	F	L	R	T	U	K
LARGE	.018	.013	.025	1/8	.045	.020	.022	.708	.013	5/16	11/32

A = WIRE SIZE

FIGURE 23—Type XVI, Class A, Point 3

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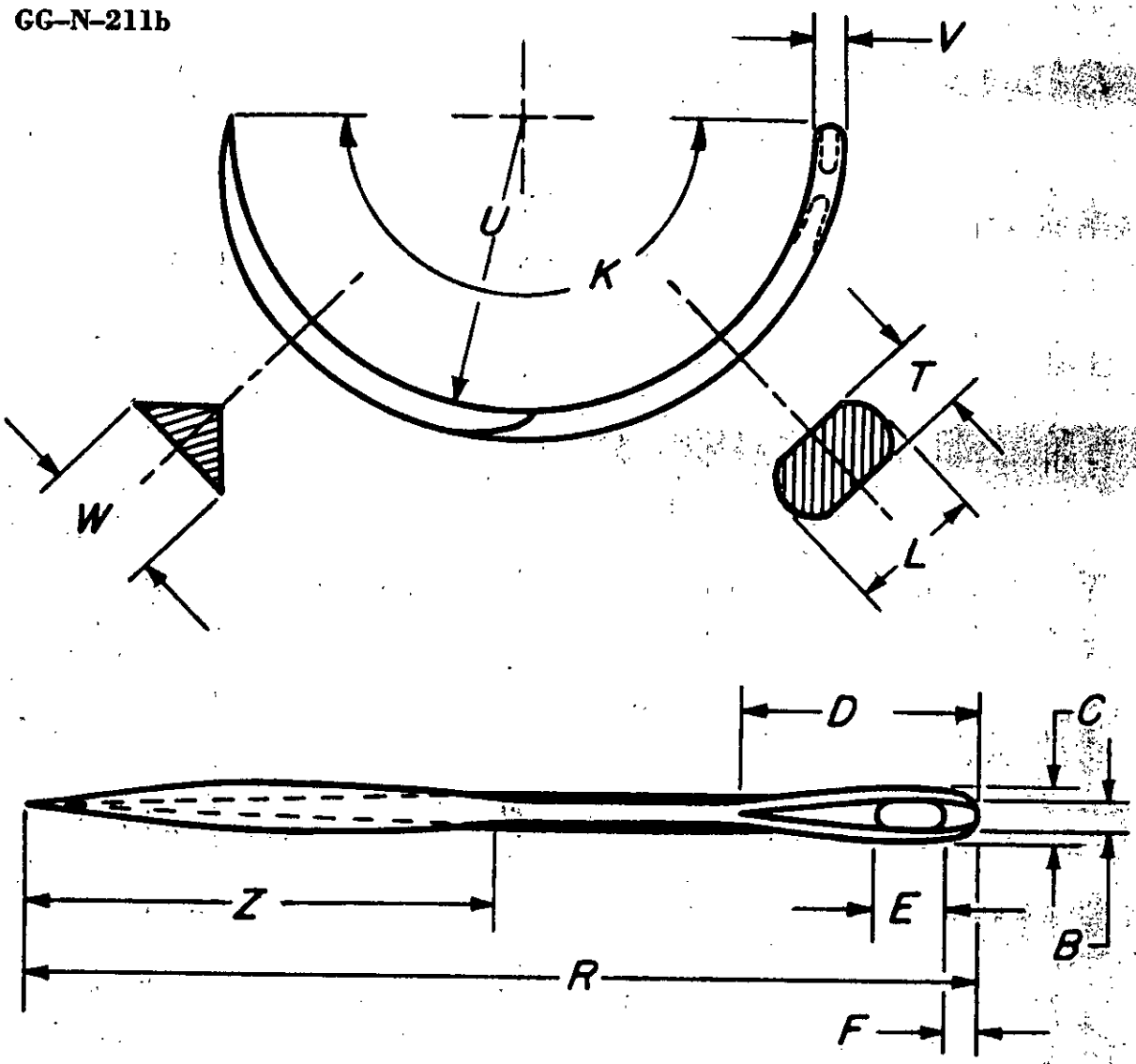


SIZE	A	B	C	D	E	F	L	R	T
1	.045	.030	.062	5/16	.090	.050	.060	3	.028

A = WIRE SIZE

FIGURE 24—Type XVII, Class D, Point 1

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NEEDLE BEFORE BENDING

SIZE	A	B	C	D	E	F	K	L	R	T	U	V	W	Z
2	.026	.022	.042	3/16	.0625	.028	1	.032	1	.024	5/16	.021	.034	3/8
3	.025	.018	.036	3/16	.050	.025	7/8	.027	7/8	.020	9/32	.016	.032	7/16
4	.025	.018	.036	3/16	.050	.025	3/4	.027	3/4	.020	15/64	.019	.032	3/8

A = WIRE SIZE

FIGURE 25—Type XVIII, Class B, Point 1

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